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GCNN: Designing efficient CNN using genetic algorithm for biosignal classification

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Abstract—In recent years, Convolutional Neural Network (CNN) architectures have achieved outstanding performance in image and time-varying signal classifications. However, the performance of CNNs highly depends upon efficient architecture, which is difficult to design manually and requires expert knowledge. This article presents an automatic efficient CNN architecture design method by using the evolutionary genetic algorithm. The method was validated with ECG datasets and proved its advantages in the design of CNN architecture. The performance of the proposed method was evaluated using different state-of-the-art CNN architectures. From the experimental results, it is clear that the proposed method has outperformed the existing methods in terms of accuracy.

Index Terms—CNN, Genetic Algorithm, ECG

1 INTRODUCTION

IoT is an emerging technology, and it can connect our surrounding physical objects with some devices having sensing and processing capability and creates a smart environment for the users [1]. IoT can be used in different applications such as agriculture, smart cities, smart traffic, healthcare, industry, etc. Smart healthcare is a critical application, and it requires intelligent devices having ultra-low latency decisions, which is very difficult in current state-of-the-art healthcare solutions [2]. Different biomedical data such as temperature, ECG signal, brain EEG signal, oxygen level, etc., are captured in smart healthcare devices. For automatic analysis of these data, an efficient analysis model is required, which can give a response in a fraction of time. Deep Learning (DL) is the most popular and widely used data analysis framework, especially for image and signal base data. However, the design of a proper DL architecture depends upon several factors, is time-consuming, and requires expert knowledge. Though the performance of the CNN is very high, designing an efficient CNN architecture is very challenging. Based on the traditional reinforcement-based designing an efficient architecture requires expert knowledge and costly computational resources [3]. To overcome this problem, evolutionary-based algorithms such as Genetic Algorithm (GA) plays an important major role in the design of a suitable efficient CNN framework with less extensive hardware resources [4], [5]. Biologically inspired evolutionary methods were utilized to enhance the overall efficiency and performance of analyzing techniques. GA is one of the most popular and widely used evolutionary approaches in solving different optimization problems [6]. GA has proved its capability by utilizing different bio-inspired operations such as selection, mutation, crossover, etc. [7].

As depicted in Figure 1, a single beat of an ECG wave is comprised of various kinds of waves called PQRST. Based on the size and pattern of these waves, it is very easy to

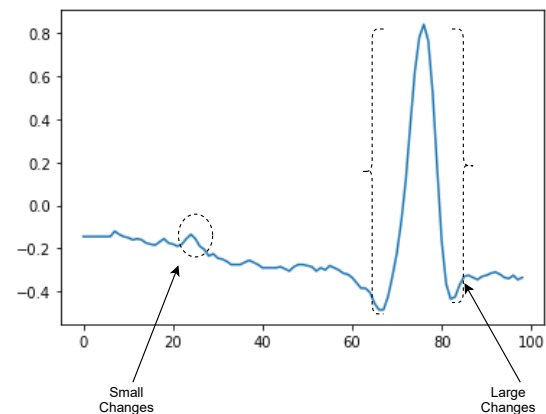


Fig. 1: Sample ECG graph with different waves

distinguish one ECG signal from others. Proper selection of accurate filter size of a CNN architecture plays a major role in extracting unique features and proper identification of ECG waves. However, accurately selecting the most appropriate filter size is a crucial process that requires a rich domain understanding. Unfortunately, not all interested people have such rich information about CNN. Therefore, the automatic selection of proper CNN parameters is very important for the efficient design of CNN architecture. Considering the inspiration mentioned above, we developed a method to construct an efficient CNN architecture using the GA, in short, called GCNN. This method automatically designs a best-performing CNN architectures using the predefined set of conditions. The novelty of the suggested GA-based CNN method is in generating the best suitable CNN architectures during the continuous evolutionary process. The main contributions of the presented GCNN method are listed as follows.

TABLE 1: Comparison of some existing GA based CNN optimization techniques

Paper	Application	Data Type	Parameters Used
Johnson et al. [8]	Optimization of the CNN architecture for image classification	Image Data	Depth, number of filters, and the kernel size
Aszemi et al. [9]	Finding best network structure	Image Data	Depth, number of filters, kernel size, learning rate, no. of hidden layers, etc.
Kilicarslan et al. [10]	Optimization of CNN parameter	Biomedical numeric data	Max-epoch, momentum, learn-rate, Batch size, optimizer, and loss, etc.
Bakshi et al. [11]	Optimisation of CNN architectures	Image Data	Number of network layers, number of feature maps, learning rate, weight decay factor, and momentum.
Gibb et al. [12]	Optimization of CNN structure	Image Data	Number of convolutional layer, filter per layer, and size of the filter
Sun et al. [13]	Designing CNN architecture	Image data	Depth of the CNN, filter size, and number of filters

- 1) Optimized the CNN architecture automatically without any proper expert knowledge.
- 2) The model is comparatively lighter than existing architectures.
- 3) The performance of the model is very satisfactory and also suitable for lightweight IoT devices.

2 RELATED WORKS

Johnson et al. [8] developed an efficient CNN architecture by optimizing using the genetic algorithm. They have optimized the information of depth, the number of filters of each convolutional layer, and the kernel size used in that layer. The fully connected section and the batch normalization and max-pooling layers are fixed. We used our algorithm to find optimal structures for the datasets MNIST, CIFAR10, and CALTECH256, running each test 10 times with different seeds for the genetic algorithm. The authors achieved impressive results in all the datasets compared to the existing algorithms. CNNs are used in image classification problems where it is necessary to generate feature descriptors to discern between image classes. CNNs by training them on images of concrete containing cracks. It is desirable to more easily find an optimal CNN structure to increase classification accuracy. In the proposed method, a GA is used to evolve the parameters that influence the structure of a CNN. Gibb et al. [12] They have optimized the information of Convolution Layers, Max-pooling Layers, Size of Convolution, and Filters Number of Convolution.

Similar to the above research article, Aszemi et al. [9] also optimized the hyperparameters of CNN architecture using the genetic algorithm. They optimized several parameters, such as the number of filters, kernel size, hidden layers, and activation value. During the experiment, the MNIST dataset was used and compared the performances in terms of accuracy with conventional CNN architecture.

Nutritional anemia is a life-threatening disease, and different researchers have used Deep Learning for the early prediction of symptoms of this disease. Kilicarslan et al. [10] mentioned that a particular analyzing method does not perform uniformly for all datasets because of several factors such as dataset sizes, quality of extracted features, etc. Finally, they proposed two hybrid models using the Genetic algorithm (GA) and deep learning algorithms of Stacked Autoencoder (SAE) and Convolutional Neural Network (CNN) for the prediction of HGB-anemia. In the proposed GA-SAE and GA-CNN models, the hyperparameters of SAE and CNN algorithms are optimized using GA since it is

not easy to determine suitable values for deep learning algorithms. Based on the different evaluation parameters, they have proved that the proposed method overcame the existing methods and achieved 98.5% accuracy.

Bakshi et al. [11] also mentioned that the performance of CNN depends on its architecture and hyperparameter settings. They developed an efficient fast-CNN architecture by optimizing hyperparameters using the genetic algorithm to analyze the CIFAR10 dataset. During the experiment, they optimized the hyperparameters like CNN layers and other hyperparameters such as the learning rate, weight decay, etc., and obtained a better performance compared to the existing manually designed CNN architectures.

Sun et al. [13] developed an automated method for the construction of CNN architecture design method for image classification by optimizing with the genetic algorithm. The developed algorithm was tested with popular and widely used standard image datasets, and its performance was compared with different state-of-the-art techniques consisting of eight manually designed CNNs, seven automatic and manually tuning, and five automatic CNN architecture design techniques. During the optimization, they used only two convolutional layers and the pooling layers, in short, they mentioned as skip layers. Each convolutional layer connect the input from the previous layer and produce output to the next layer.

3 PROPOSED METHODS

The proposed architecture for efficient auto-selection of the hyperparameters of a CNN architecture is shown in Fig. 2. In the resource-constrained device, the implementation of a complex CNN architecture is not suitable. To bypass this problem Genetic Algorithm was used for the proper selection of hyperparameters to make a CNN architecture more efficient. In the proposed method the size of the fully connected layer remains fixed only the parameters such as convolutional filter size, number of filters, padding size, and stride size are optimized using the Genetic algorithm. The proposed GCNN method passes through a series of evolutionary processes to discover the best architecture. During the evolutionary process of GA, fitness of each individuals of a populations are estimated. During fitness evaluation, the system automatically encodes a CNN architecture and tested with a given dataset. In this experimnt classification accuracy is used to measure the fitness value. Using this fitness value parent individuals are selected and performed the crossover and mutation operation to generate the new generation. After generating the offspring, individual for next generation are selected based on the fitness value from parent and offspring generation. An in-depth explanation of all the important steps of the proposed technique is given below.

3.1 Population Generation

In this research we are trying to find out a suitable CNN architecture for the analysis of ECG signal. The performance of CNN is highly depends upon the extraction of quality features, which is relies on the convolutional layer. An ECG signal consist of different types of waves, and

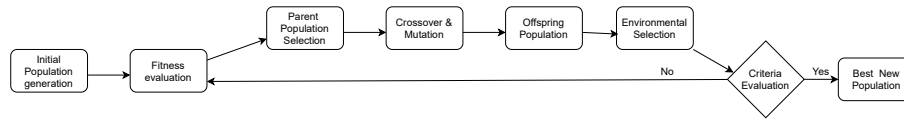


Fig. 2: Flow chart of the proposed method

requires different filter size for proper extraction of features. To handle this issue, genetic algorithm was used to find out the proper combination of filter size and number of filters to design a suitable CNN model. In this population initialization phase, we encoded the building blocks of a CNN architectures. Initially, N numbers of individuals are generated randomly having same length, same number of convolution and pooling layers. During initialization, each individual have different number of convolutional filters with different filter size.

Algorithm 1 Estimation of Individual Fitness Value

$P_i \leftarrow$ Set of Individual

$F_p \leftarrow$ Features Set

$A_p \leftarrow$ Accuracy Set

Input:

$P \leftarrow$ Population P containing Individual

$E \leftarrow$ ECG signal dataset

Output:

$F_{ip} \leftarrow$ Fitness level of all individuals

- 1: **procedure** FEATURE EXTRACTION(P_i)
- 2: Auto setup a CNN model based on the encoding information
- 3: Using the constructed model extract features automatically from the ECG datasets stored in F_p
- 4: **procedure** CALCULATE ACCURACY(F_p)
- 5: Divide the feature set in 80:20 ratio
- 6: Train the Classifier
- 7: Calculate the classification accuracy A_p
- 8: **procedure** AVERAGE ACCURACY(A_p)
- 9: Compute average accuracy F_{ip} from the set A_p
- 10: **for** each and every member in a population **do**
- 11: $F_p = \text{FeatureExtraction}(P_i)$
- 12: $A_p = \text{CalculateAccuracy}(F_p)$
- 13: $F_{ip} = \text{AverageAccuracy}(A_p)$

3.2 Fitness Evaluation of Each Individual

The fitness evaluation of an encoded CNN architecture is noted in the Algorithm 2. In this process, for a given dataset fitness of all the individuals of a population is evaluated. During this process, from the encoded information of an individual a CNN architecture is constructed. Each constructed CNN models are trained with a given dataset and stored its detection accuracy. For training and evaluating the detection accuracy softmax classifier was used.

3.3 Offspring Generation

As mentioned in the Algorithm 2, initially two individuals called as parents are selected having highest fitness value.

Algorithm 2 Offspring Generation

$P \leftarrow$ Parents

Input:

$P \leftarrow$ Set of all individuals with fitness value

$P_c \leftarrow$ probability of crossover

$P_m \leftarrow$ Probability of mutation

Output:

$O \leftarrow$ New offspring generation

- 1: **procedure** PARENTS SELECTION(P_i)
- 2: Randomly select two individuals P_1 and P_2 having the highest fitness value
- 3: **procedure** CROSSOVER(P_1, P_2)
- 4: Choose a crossover point randomly and divide the individual encoded information into two parts
- 5: $O_1 \leftarrow \text{Join the first part of } P_1 \text{ and second part of } P_2$
- 6: $O_2 \leftarrow \text{Join the first part of } P_2 \text{ and second part of } P_1$
- 7: $Q_t = O_1 U O_2$
- 8: **procedure** MUTATION(P_i)
- 9: Randomly choose the mutation point in P
- 10: Compute the mutation
- 11: $r =$ randomly generate a number between 0 and 1
- 12: **while** ($r \leq P_c$) **do**
- 13: $C_p = \text{Crossover}(P_1, P_2)$
- 14: $r =$ randomly generate a number between 0 and 1
- 15: **while** ($r \leq P_m$) **do**
- 16: $M_p = \text{Mutation}(P_i)$
- 17: Return Q_t

During the crossover operations both the parents are randomly splitted into two parts and swapped to generate the new individuals. The number of convolutional and pooling layers plays a crucial role in the extraction of important features which affect the performance of a CNN architecture. During the mutation operation, initially, a random number is generated and compared with the probability of mutation. If the random number value is less than or equal to the probability of mutation then the mutation operation is performed. In this method during the mutation operation, the number of filters of a convolutional layer was increased to improve the quality of extracted features.

3.4 Environmental Selection

This process basically select the best individual for the upcoming generation. In this process individual are considered from both the parent and offspring population. As mentioned in the Fig 4, individuals with maximum fitness value are considered for the upcoming generation.

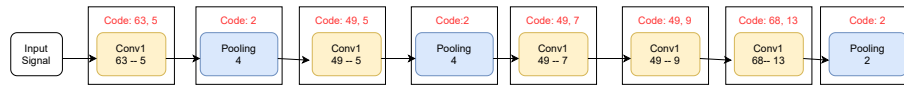


Fig. 3: Representation of a CNN in encoded form. So the code representing this CNN architecture is: 63-5-2-49-5-2-49-7-49-9-68-13-2

Algorithm 3 Offspring Selection for Next Generation

Input:

$P_t \leftarrow$ Set of parent individuals

$P_s \leftarrow$ Size of the parent population

$Q_t \leftarrow$ offspring population

Output:

$P_{t+1} \leftarrow$ The population for next generation

- 1: **while** ($P_{t+1} < P_s$) **do**
- 2: Randomly select two individual P_1 and P_2 from $Q_t \cup P_t$
- 3: $R =$ Select the one with better fitness value from P_1, P_2
- 4: $P_{t+1} = P_{t+1} \cup R$
- 5: Find the individual with the best fitness from P_{t+1}

TABLE 2: Comparison of the proposed method GCNN with different conventional CNN architecture

Method	ACC	FP	FN
LeNet	85	9	6
AlexNet	94	3	3
VGG-16	93	2	5
VGG-19	93	2	5
Inception	92	4	4
ResNet	89	5	6
DenseNet	90	3	7
Ensemble	96	2	2
GA-CNN	98	1	1

- 4) During the crossover of an individual, the center point is selected as the crossover location.
- 5) The performance of the proposed method is compared with other state-of-the-art methods using the accuracy and the execution time.

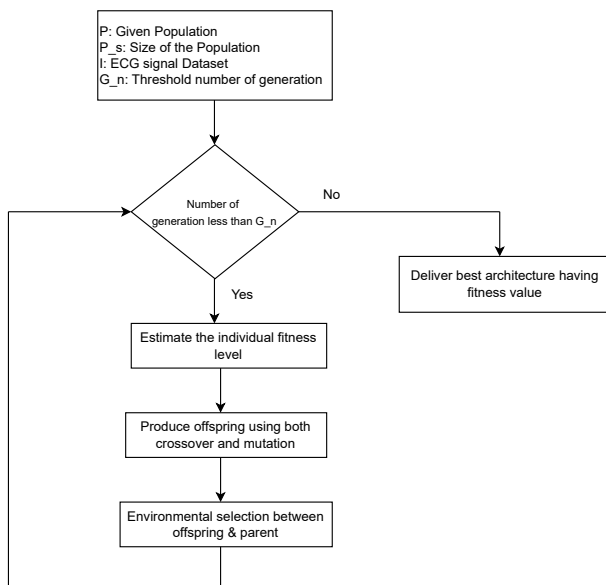


Fig. 4: A simple ECG graph with different changes

4 EXPERIMENTAL SETUP

To perform this experiment we have followed the following steps.

- 1) Here, to perform the experiment we have used an ECG dataset containing 1500 signals.
- 2) The initial architectures were generated based on the LeNet-5 model, and we set 8 number of individuals in the population.
- 3) The experiment is performed for 14 number of generation.

5 RESULTS

In this experiment, to better understand the performance of the proposed method, we have set a total 14 number of generations. The evolutionary graph of the proposed method is shown in Fig 5. As depicted in Fig 5, the classification accuracy of the proposed GCNN with the given dataset is increasing slowly. It can be observed that the classification accuracy is minimal in the first generation and it increases till the 11th generation. After the 11th generation, the improvement in the accuracy is very minimal almost negligible.

Table 2. also presents a comparison of different conventional CNN algorithms with the proposed GCNN method. From all the presented results it is clear that the proposed GCNN method achieved a better accuracy compared to the other methods after eleven iterations. As mentioned in the experimental setup, the GCNN method basically used the conventional LeNet-5 method and increased its performance from 85% to 98%.

6 CONCLUSION

In critical applications, quick response is necessary and demands a lightweight analysis model. We can easily design an efficient compact CNN architecture where the number of operations will be minimal. To achieve this we used the bio-inspired evolutionary approach called as Genetic Algorithm. Using this evolutionary approach the performance of the conventional LeNet-5 model has increased from 85% to 98%. This achievement has proved the ability to use an evolutionary approach with the conventional CNN method. In the near future, we will try to use the GA method with different Conventional and hybrid CNN techniques.

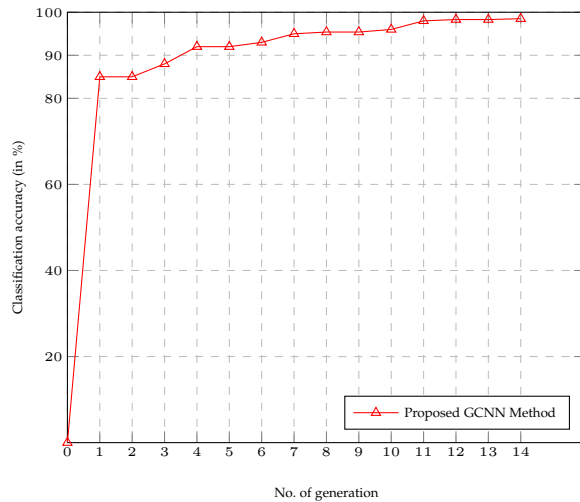


Fig. 5: Evolution trajectory of the proposed algorithm

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